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Clinical Biomechanics 13 (1998) 584–592

CLINICAL BIOMECHANICS

The effect of imposed and self-selected computer monitor height on posture and gaze angle

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Received 22 September 1997; accepted 3 March 1998

Abstract

Objective: The objectives were to determine the postural consequences of varying computer monitor height and to describe self-selected monitor heights and postures.

Design: The design involved experimental manipulation of computer monitor height, description of self-selected heights, and measurement of posture and gaze angles.

Background. Disagreement exists with regard to the appropriate height of computer monitors. It is known that users alter both head orientation and gaze angle in response to changes in monitor height; however the relative contribution of atlanto-occipital and cervical flexion to the change in head rotation is unknown. No information is available with regard to self-selected monitor heights.

Methods. Twelve students performed a tracking task with the monitor placed at three different heights. The subjects then completed eight trials in which monitor height was first self-selected. Sagittal postural and gaze angle data were determined by digitizing markers defining a two-dimensional three-link model of the trunk, cervical spine and head.

Results. The 27° change in monitor height imposed was, on average, accommodated by 18° of head inclination and a 9° change in gaze angle relative to the head. The change in head inclination was achieved by a 6° change in trunk inclination, a 4° change in cervical flexion, and a 7° change in atlanto-occipital flexion. The self-selected height varied depending on the initial monitor height and inclination.

Conclusions. Self-selected monitor heights were lower than current 'eye-level' recommendations. Lower monitor heights are likely to reduce both visual and musculoskeletal discomfort.

Relevance

Musculoskeletal and visual discomfort may be reduced by placing computer monitors lower than currently recommended.
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Keywords: VDT; Posture; Head and neck

1. Introduction

Disagreement exists with regard to the appropriate height of computer monitors. The orthodox view is that the monitor should be located at, or just below, eye level. For example, the Australian Occupational Health and Safety Commission suggests in a guidance note that 'when sitting tall and looking straight ahead, the keyboard user should be looking at the top edge of the screen' [1]. Similarly, the Canadian Standards Association recommends that 'the top of the screen should be at approximately the operator's eye level when the head is held up' [2].

Such recommendations are not based on empirical evidence, and have been challenged. Ankrum and Nemeth [3] argued that the top of the monitor should be located at least 15° below horizontal eye level; while Kroemer *et al.* [4] suggested 30° or more below horizontal eye level.

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0268-0033/98/$19.00 + 0.00 © 1998 Elsevier Science Ltd. All rights reserved.
PII: S0268-0033(98)00021-7

The argument for lower monitor placement is based on the observation that the subjective preference is for visual targets to be located such that the eyes are rotated downwards relative to the head [5–8]. Kroemer and Hill [9], for example, measured the average preferred gaze angle as 35° below the Ear–Eye Line for visual targets at 1 m.

Jampel and Shi [10] reported that the Ear–Eye Line is typically about 15° above the horizontal in an erect posture, while Jones *et al.* [11] reported that the orientation of the head in the most comfortable sitting posture corresponded to an approximately horizontal Ear–Eye Line (although considerable individual variability existed). These observations suggest that locating video monitors at the orthodox 'eye level' position is likely to require users to either compromise their preferred gaze angle or to adopt postures that may cause discomfort.

The argument for lower computer monitors is consistent with epidemiological research that has found an association between 'eye level' monitor heights and neck discomfort [12]. However, it relies on an assumption that when monitors are lowered users do not rotate the head anteriorly by the same angular extent as the change in gaze angle relative to the horizontal.

To put it another way, users may potentially respond to a lower monitor location in three ways: (1) posture may remain constant while gaze angle relative to the head lowers; (2) the gaze angle relative to the head may be maintained by rotating the head anteriorly through some combination of trunk, cervical, or atlanto-occipital flexion; or (3) both head orientation and gaze angle relative to the head may alter. Previous research has consistently supported the third alternative [13–16]. Changes in the orientation of the head appear to occur through a combination of flexion at cervical and atlanto-occipital joints. However, the relative contribution of different joints is unknown. This information is necessary to evaluate the likely biomechanical consequences of lower monitor locations.

A consequence of increased cervical and/or atlanto-occipital flexion is an increase in the flexor moment caused by the mass of the head, and head and neck, about axes of rotation at the level of the atlanto-occipital joint and the cervical spine; and consequently greater tension must be provided by the musculature to maintain static equilibrium. However, Kumar [15] found that, even though the calculated flexor moment of the head and neck about C7 was increased with lower monitor placement, trapezius activation and subjective ratings of discomfort were both decreased. Reconciliation of these apparently paradoxical results, and understanding of the functional consequences of the postural changes, requires knowledge of the relative contributions of different joints to the changes in head orientation adopted in response to different visual target locations. The aim of the first part of this experiment is to provide this information by measuring the changes in trunk, cervical and atlanto-occipital postures (as well as the changes in head inclination, and gaze angle relative to the head) that occur when gaze angle relative to the horizontal is altered by changing monitor height.

Given that users alter head orientation to some extent when the vertical location of a visual target changes, it is not possible to specify the range of appropriate monitor locations necessary to accommodate individual differences solely on the basis of information about preferred gaze angle relative to the head. The aim of the second part of this experiment was to address this question by asking subjects to select the monitor height that they believe is most comfortable, and then to determine the postures and gaze angles adopted to view these monitor locations. Specifically, the aims were to determine what range of monitor positions (defined in terms of gaze direction relative to the horizontal) are selected by users when the vertical location of the computer monitor may be continuously adjustable through a large range; and to describe the postures that are adopted to view a monitor placed in these self-selected locations.

2. Methods

2.1. Subjects

Six female and six male university students aged 19 to 49 (mean = 26 years) volunteered to participate. All had normal or corrected vision. All participants used computers in their daily activities, but none were employed full-time in occupations that involved computer use.

2.2. Procedure

The task used was a computer game that required targets to be tracked using a mouse-controlled cursor. The task was chosen to avoid potential confounding by requiring visual fixation on the screen only. The workstation consisted of a height- and tilt-adjustable chair, a height- and tilt-adjustable monitor (viewing area 255 × 190 mm), and a table (height = 690 mm) on which a mouse pad and mouse were placed. Each subject participated in four trials (5 min each) where the monitor location was imposed (part 1), followed by a series of eight trials (two in each of four initial location conditions) in which the subject first selected a monitor location that was perceived to be most

comfortable, and then performed the task for 5 min (part 2). All trials were completed on the same day.

The four imposed trials were performed at three monitor locations (high, middle and low). An attempt was made to standardize eye position by adjusting the subjects' chair height such that, when sitting upright and looking straight ahead, the subjects' eyes were all at a constant height of 1.24 m. The subjects were, however, free to move during each trial (although no adjustment of the chair was permitted) and consequently some variability existed in the resulting gaze angle with respect to horizontal. Each subject performed two trials with the monitor in a high location. In these trials the average gaze angle to the centre of the screen was 2° below the horizontal. In the middle and low monitor height conditions, the average gaze angles were, respectively, 15 and 29° below horizontal. Eye to screen distance remained approximately constant across conditions (mean = 0.87 m). Screen inclination was adjusted to remain approximately perpendicular to the line of gaze. Each trial was 5 min in duration and the order was counter-balanced across subjects.

Each subject then completed eight trials in which the monitor height and inclination were self-selected. Prior to the adjustment of monitor height, the monitor was either set to vertical or inclined backwards by 40°, and lowered or raised to the extremes of the adjustment mechanism. In the ascending conditions, the monitor screen centre began at 60 cm above floor height and was slowly raised by the experimenter until the subject indicated that the current height was most comfortable. In the descending conditions, the monitor screen centre began at 130 cm above the floor and was slowly lowered until the subject indicated that the current height was most comfortable. After subjects selected their preferred monitor height for each trial, the preferred monitor inclination was selected in a similar fashion. Subjects then performed the task for a 5-min period. Each of these four initial monitor locations (high 0, high 40, low 0, low 40) was presented to each subject twice in one of two counter-balanced and pseudo-randomized orders.

Spherical reflective markers were attached to the outer canthus of the eye (OC), the mastoid process on a line joining the tragus and the outer canthus (MP), the spinous process of the seventh cervical vertebra (C7) and the greater trochanter (GT). These markers were used to define the position of the head and neck in the sagittal plane. The head and neck were modelled as three rigid links articulated at pin joints located at the level of the atlanto-occipital joint and between C7 and T1 (see Fig. 1). The markers remained attached for the duration of the experiment.

The position of the head relative to the environment was described by inclination of the line joining MP and OC markers (the Ear–Eye Line) with respect to the horizontal. Sagittal movement of the head about the atlanto-occipital joint was described by the anterior angle subtended between C7, MP and OC markers (termed head angle). Sagittal movement of the neck about the cervical joints was described as the anterior angle subtended by MP, C7 and GT markers (termed neck angle). The orientation of the trunk relative to the environment was described as the inclination of a line joining C7 and GT markers relative to the horizontal. The gaze angles to the monitor centre relative to the horizontal, and relative to the orientation of the head, were calculated as defined in Fig. 1 via a marker placed at the screen centre.

The reflective markers were illuminated by a 1000-W light placed behind the camera, and the movement of these markers recorded on a Panasonic AG-6300 VHS recorder using a NEC TI-23A CCD camera. The two-dimensional joint angular kinematics were subsequently obtained via automated digitisation (at 30 Hz) of these markers. The tape was replayed through a video processor (VP110, Motion Analysis Corporation, California, USA) and the two-dimensional coordinates of each marker were obtained by calculating the centroid of each marker outline for each digitized video frame. Standard FLEXTRAK software was used to generate two-dimensional spatial paths as a function of time for each of the four markers.

Each trial included a 5-min work phase in which one sample of 50 s was collected from the last minute. For each data point, the position of the Ear–Eye Line with respect to the horizontal, trunk inclination, head, neck and gaze angles were calculated. These data were filtered at 2 Hz (Butterworth recursive, low pass) and reduced by extracting every tenth frame, resulting in a 150-point time series for each angle (an effective frame rate of 3 Hz).

2.3. Analysis

Mean values were calculated for the 50-s sample taken from each trial. One-way analysis of variance (ANOVA) and 95% confidence intervals were used to describe the effect of imposed monitor height conditions on the posture and gaze angles adopted. Two-way (initial height × initial inclination) ANOVA and 95% confidence intervals were used to describe the postures and gaze angle adopted in self-selected monitor height conditions and the effect of initial starting height and monitor inclination on these postures and gaze angles. Subject mean values across imposed conditions were also correlated with the corresponding values across self-selected condition using Pearson product moment correlation coefficient (r) and 95% confidence intervals were calculated using a Fisher z-transformation.

3. Results

3.1. Imposed monitor heights

Subjects responded to changes in monitor heights by changing both head orientation (as measured by Ear–Eye position) and the gaze angle relative to the head (gaze angle relative to Ear–Eye Line). A 27° change in gaze angle relative to the horizontal imposed by the changes in monitor location was achieved, on average, by an 18° change in head orientation (Ear–Eye position) and a 9° change in gaze angle relative to the Ear–Eye Line.

The 18° change in head orientation from high to low monitor locations was achieved by increased flexion of the trunk, cervical, and atlanto-occipital joints (6° average change in trunk inclination, 4° change in neck angle, and 7° change in head angle; see Fig. 2 and Table 1). Examination of the Table 1 and Fig. 2 revealed that the largest change occurred at the atlanto-occipital joint (as measured by head angle), and that inter-subject variability increased with lower monitor height. As the monitor was lowered from 2 to 29° below horizontal, the average gaze angle to the monitor centre changed from 21 to 30° below the Ear–Eye Line.

3.2. Self-selected monitor heights

Considerable individual differences existed in the monitor heights selected by the subjects as being most comfortable. Gaze angles ranged from 6° above horizontal to 42° below horizontal, and gaze angles relative to the head ranged from 9 to 53° below the Ear–Eye Line.

Self-selected monitor height was influenced by the initial height and inclination of the monitor (Table 2, Fig. 3). Two-way (initial monitor height × initial inclination) ANOVA revealed that all two-way interactions were non-significant ($P > 0.05$), which indicates that the effects of initial monitor height and inclination on the monitor height selected (and posture adopted at

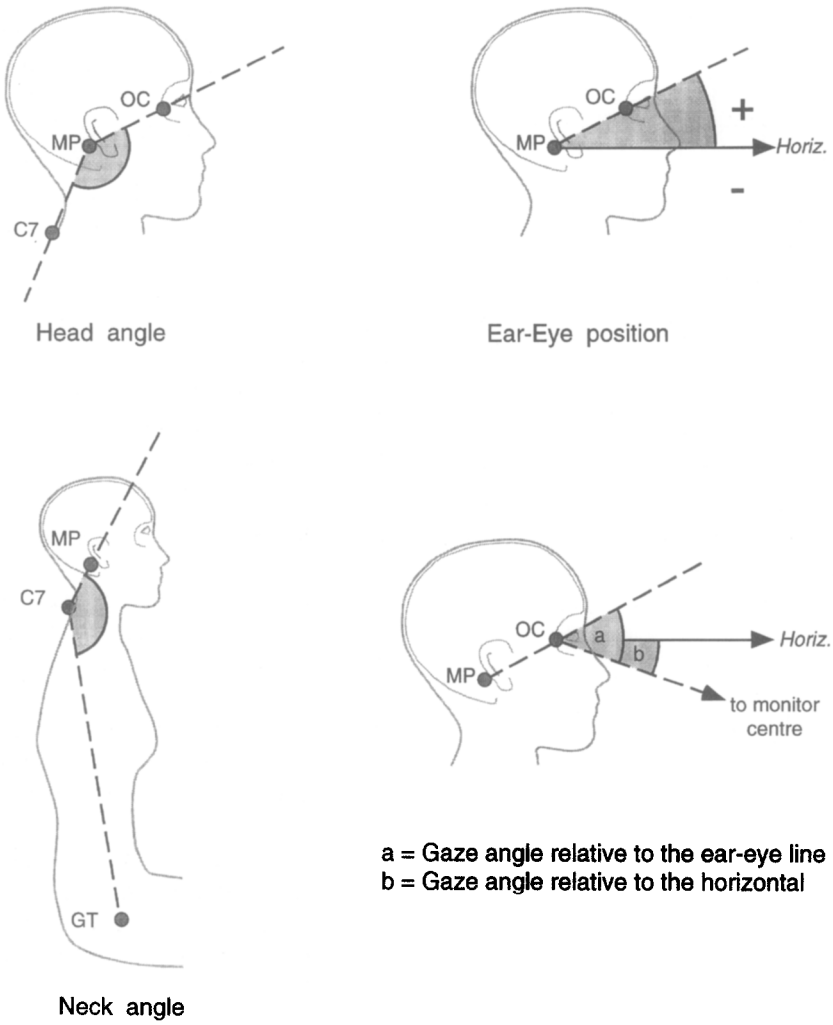


Fig. 1. Marker placements and definition of angles.

these positions) were independent. There were, however, significant main effects of both initial monitor height and inclination.

When the initial monitor location was low, the average monitor height selected by subjects resulted in a gaze angle 9° lower relative to the horizontal

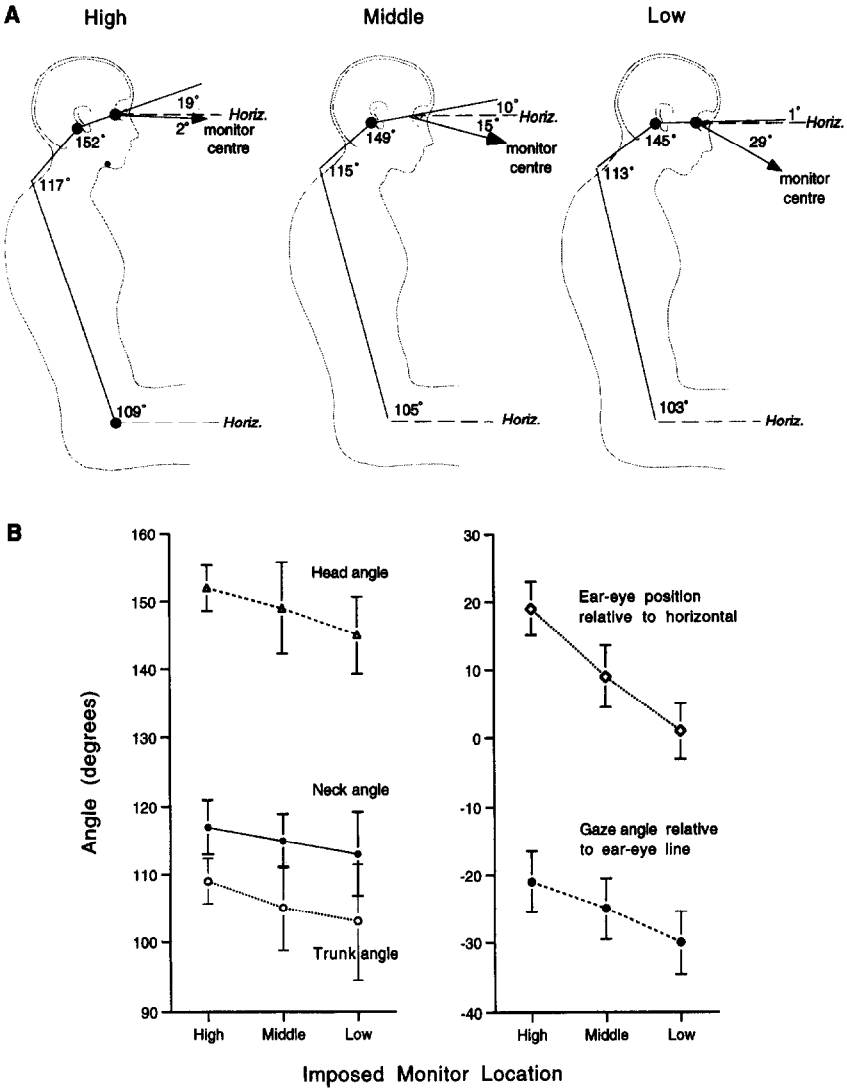


Fig. 2. Schematic presentation of average postures adopted in imposed monitor height conditions (A) and mean values with 95% confidence intervals for these data (B).

Table 1
Means (95% confidence intervals) and ANOVA summary statistics for measurements describing the postures and corresponding gaze angles adopted in the imposed monitor height conditions

Measurement (deg)	High monitor	Middle monitor	Low monitor	F	P
Gaze angle relative to horizontal	−2 (±1)	−15 (±1)	−29 (±2)	998	<0.001
Ear–Eye Line relative to horizontal	19 (±4)	9 (±5)	1 (±4)	200	<0.001
Gaze angle with respect to Ear–Eye Line	−21 (±5)	−25 (±5)	−30 (±5)	46	<0.001
Head angle	152 (±3)	149 (±7)	145 (±6)	8.4	= 0.002
Neck angle	117 (±4)	115 (±4)	113 (±6)	3.1	= 0.064
Trunk orientation relative to horizontal	109 (±3)	105 (±6)	103 (±9)	1.7	= 0.20

Note: All degrees of freedom for ANOVA = (2, 22).

Table 2
Means (and 95% confidence intervals) for measurements describing the postures and corresponding gaze angles adopted in the self-selected monitor height conditions

Measurement (deg)	High 0	High 40	Low 0	Low 40
Gaze angle relative to horizontal	−3 (±3)	−11 (±7)	−12 (±5)	−20 (±7)
Ear–Eye Line relative to horizontal	19 (±4)	14 (±5)	12 (±5)	8 (±5)
Gaze angle with respect to Ear–Eye Line	−22 (±5)	25 (±5)	−24 (±5)	−27 (±6)
Head angle	152 (±4)	151 (±5)	149 (±5)	147 (±5)
Neck angle	116 (±5)	114 (±4)	114 (±4)	112 (±5)
Trunk orientation relative to horizontal	111 (±6)	110 (±5)	110 (±5)	109 (±6)

($P = 0.04$). Similarly, when the monitor was initially inclined backwards by 40° the average monitor height selected by subjects resulted in a gaze angle 8° lower relative to the horizontal ($P = 0.001$); and these effects were additive.

Significant main effects of initial height were observed for head angle ($P = 0.002$) and gaze angle relative to Ear–Eye Line ($P = 0.04$); significant effects of initial inclination were observed for neck angle ($P = 0.005$) and gaze angle relative to Ear–Eye Line ($P = 0.005$). All other main effects were not statistically significant ($P > 0.05$).

While individual differences existed between subjects in the self-selected monitor locations and postures adopted, there was relatively less variability within the subjects across imposed and self-selected trials. Significant correlations between subject average values in imposed and self-selected conditions existed for head angle ($r = 0.93$, 95% confidence interval 0.76 to 0.98), neck angle ($r = 0.87$, 95% confidence interval 0.59 to 0.96) and gaze angle relative to the head ($r = 0.94$, 95% confidence interval 0.79–0.98). Trunk posture was not significantly correlated ($r = 0.20$, 95% confidence interval −0.42 to 0.69).

4. Discussion

The postures adopted to view visual targets such as computer monitors are a consequence of an interaction between the characteristics of the visual and musculoskeletal systems. This conclusion is justified by the

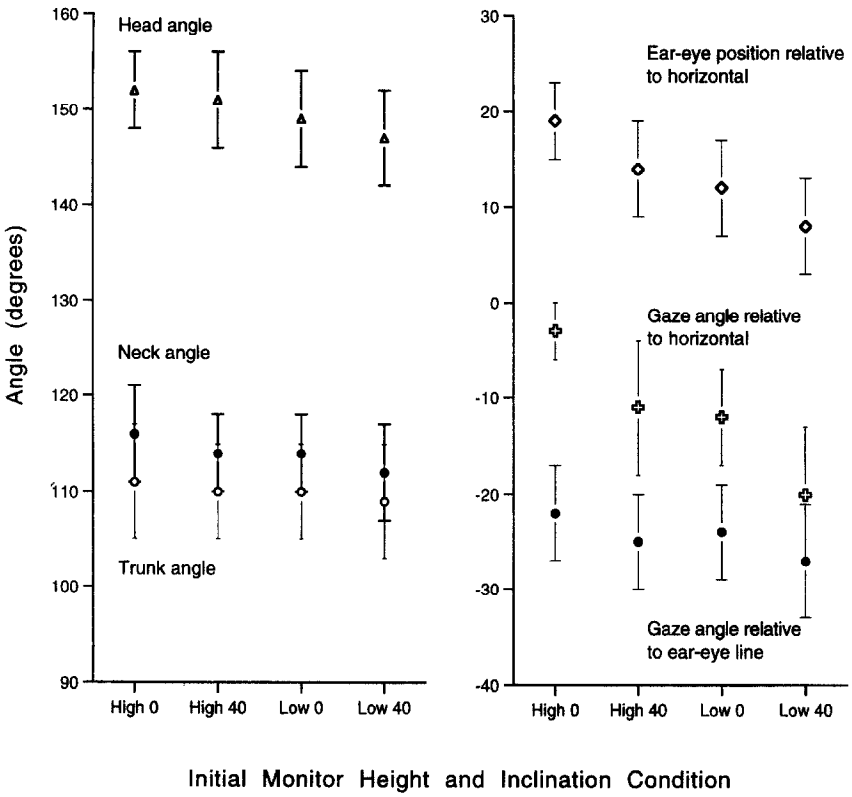


Fig. 3. Self-selected monitor height, and head and neck posture, as a function of initial monitor height and inclination.

observation (consistent with previous research) that changes in monitor height are accompanied by changes in both head inclination and gaze angle relative to the head.

The orientation of the Ear–Eye Line in the imposed high monitor condition was 19° above horizontal, which suggests that the average posture adopted involved greater head and neck extension than either the erect posture defined by Jampel and Shi [10] (Ear–Eye Line 15° above horizontal) or the most comfortable posture reported by Jones *et al.* [11] (Ear–Eye Line horizontal).

In the high monitor condition the average gaze angle was 21° below the Ear–Eye Line. When the monitor height was lowered, the average gaze angle relative to the head lowered to 30° below the Ear–Eye Line. Comparison of these values with the average preferred gaze angles determined in previous research [9] (35° below Ear–Eye Line at 1 m distance) suggests that, in all conditions, subjects typically adopted postures that did not achieve their preferred gaze angle. In the 'low' condition the gaze angle relative to the head was more likely to be closer to preferred.

The individual variability in self-selected monitor heights is consistent with the variability noted in previous experiments investigating preferred gaze angles [7,9]. The effect of initial monitor inclination is also consistent with previous research [17].

An interpretation of the functional consequences of these observations requires consideration of the biomechanics of the head and neck. The head and neck system comprises a rigid mass (the head) located above a relatively flexible cervical spine. Flexion and extension are possible at the atlanto-occipital and cervical joints. The ligaments and joint capsules are relatively elastic, especially within the mid range, and a large range of movement is possible without significant contribution from passive tissues [18].

The centres of mass of the head, and the head and neck combined, are anterior to the atlanto-occipital and cervical joints. Consequently, when the trunk is vertical, extensor torques about the atlanto-occipital and cervical joints are required to maintain static equilibrium. A large number of muscles with diverse sizes, characteristics and attachments are capable of contributing to these torques. The suboccipital muscles, which take origin on C1 and C2 and insert on the occipital bone, are capable of providing extensor torque about the atlanto-occipital joint only; others (such as semispinalis capitis and cervicus) provide extensor torque about cervical as well as atlanto-occipital joints; while others (such as iliocostalis cervicus) provide extensor torque about cervical vertebrae only [19,20]. While the sternocleidomastoids have been proposed as extensors of the atlanto-occipital joint and flexors of the cervical spine [21], their attachments on the mastoid processes are very close to the axis of rotation, and consequently the sternocleidomastoids provide no significant extension torque about the atlanto-occipital joint [22].

Increased flexion at the atlanto-occipital joint increases the horizontal distance of the centre of mass of the head from its axis of rotation (level with the mastoid process). Similarly, with the trunk in a vertical position, an increase in flexion of the cervical spine increases the horizontal distance of the combined head and neck centre of mass from the axes of rotation in the vertebral column (and all else remaining the same, the horizontal distance of the head from its axis of rotation). Hence, with the trunk approximately vertical, both atlanto-occipital and cervical flexion increase the torque required of the extensor musculature to maintain static equilibrium.

Static postures involving neck flexion beyond 30° are associated with decreased time to isometric fatigue [23], presumably a consequence of the increased load moment. According to one model [24], neck *extension* of 30° places the centres of mass approximately over the axes of rotation and reduces the external flexor moment required to resist gravitational acceleration to zero. In contrast to the suggestion made by Ankrum and Nemeth [3] and Kroemer *et al.* [4], this logic has prompted a recommendation [25] to *increase* the height of visual targets such as computer monitors in order to increase neck extension and reduce muscular effort.

The argument that muscular effort will be reduced by locating computer monitors at, or above, eye level is unconvincing because the model from which the conclusion is derived is not sufficiently complex to capture the behaviour of the system in this situation. The head and neck complex is inherently unstable, especially in the upright position [22]. The neck muscles must do more than just balance the external forces acting on the system. For the system to be stable, additional co-contraction is required to increase the stiffness of the cervical spine and prevent buckling. The consequence is that significant muscular activity is probably still required even if the head and neck are positioned to minimize the flexor torque imposed by gravitational acceleration. Indeed, the necessity for muscle activity to stabilize the cervical spine is likely to be greater when it is relatively extended [22].

In general, changes in posture at the atlanto-occipital and cervical joints will alter both the moment arm and the average fibre length of the muscles that are active to provide the required extensor torque and stiffness. While accurate estimates of moment arm and fibre length changes are available for only a few of the more than 40 muscles of the neck, it is clear that muscle fibres that produce extensor torque will be shortened by increased extension of the head and neck.

What is open to speculation is the gradient of the length/tension relationship at this point.

The above contributes to an understanding of the mechanism by which postures involving a combination of flexion at cervical joints and extension at the atlanto-occipital joints (a forward head posture [26]) may lead to discomfort. While not well quantified, such postures have been associated with physical complaints such as headaches [27,28].

Maintenance of a forward head posture places the centre of mass of both head, and head and neck combined, anterior to the respective centres of rotation, thus requiring extensor moments about both atlanto-occipital and cervical joints. Such a posture involves a reduction in the average fibre length of the muscles that contribute to the necessary extensor torque about the atlanto-occipital joint, and possibly the average fibre lengths of some of the muscle fascicles that contribute to the co-contraction necessary to stabilize the cervical spine in this position. While the length/tension characteristics of individual fascicles are unknown, it is probable that this shortening reduces the tension generating capabilities of these muscles. Given that the cervical spine is flexed, the possible contribution to atlanto-occipital extensor moment by muscles that extend the cervical vertebra is reduced and, consequently, suboccipital muscles contribute to atlanto-occipital extension moment. These muscles are relatively short, and even a small change in average fibre length caused by extension of the atlanto-occipital joint is likely to cause significant decrement in their tension generating capabilities. Yet it is precisely these muscles that appear to be primarily responsible for vertical movements about axes high in the cervical spine [22].

While changes in the imposed monitor location resulted in changes in all angles measured, the largest change was decreased atlanto-occipital extension at lower monitor heights. This response has the consequence of lengthening the muscles of the neck that insert on the skull (such as semispinalis capitis and cervicus, and the suboccipital muscles). These muscles contribute to both the extensor moment necessary to balance the flexor moment caused by the mass of the head and neck, and the stiffness necessary to maintain stability of the cervical spine and head system. Although the gradient of the length–tension relationship in each muscle in this posture is not known, lengthening of these muscles is likely to be functional in reducing the muscular effort required, despite the increase in resultant joint moment which necessarily accompanies an increase in head and neck flexion. This analysis provides an explanation for the apparently paradoxical finding made by Kumar [15] that electromyographical amplitude of the trapezius (which also takes origin on the occiput) and subjective ratings of discomfort, were both decreased with lower monitor locations. (The contribution by passive connective tissues suggested by Kumar is unlikely because a large range of movement at the head and neck occurs before any significant resistance from passive tissues occurs [18].)

Individual differences in visual and musculoskeletal systems make it unlikely that a single monitor height exists that is optimal for all users. These differences may be the source of the individual differences observed in monitor heights selected as subjectively most comfortable. Given the high correlation across imposed and self-selected conditions in head and neck postures (and gaze angles relative to the head), it is unlikely that the pattern of differences was random. The differences are more likely to reflect individual differences in characteristics of the visual and musculoskeletal subsystems. These individual differences should be reflected in the appropriate standards and documentation destined for users. These differences also imply that the optimal design of furniture should include an appropriate range of monitor height adjustability.

There are too many unknowns for a definitive conclusion to be reached with regard to an optimal monitor location. It is not clear, for example, whether the results would hold for different sized monitors, different tasks, or users of different experience. However, when the data presented here are considered in conjunction with the characteristics of the visual system and epidemiological data, the weight of evidence is such that recommendations of the 'top of monitor at or slightly below eye height' type must be questioned. Recommendations must recognize the range of individual variability, and that lower monitor locations may allow users to achieve gaze angles closer to those preferred, while adopting postures that do not involve relatively extended cervical or atlanto-occipital joints.

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